Digital Forensics (IY5609 / IY4609 / IY3609)
Unit 1 – Introduction
Welcome & Administrative Matters

Dr Olga Angelopoulou

Academic Year 2025/26

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Reminder: Record your Attendance!

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Administrative Matters

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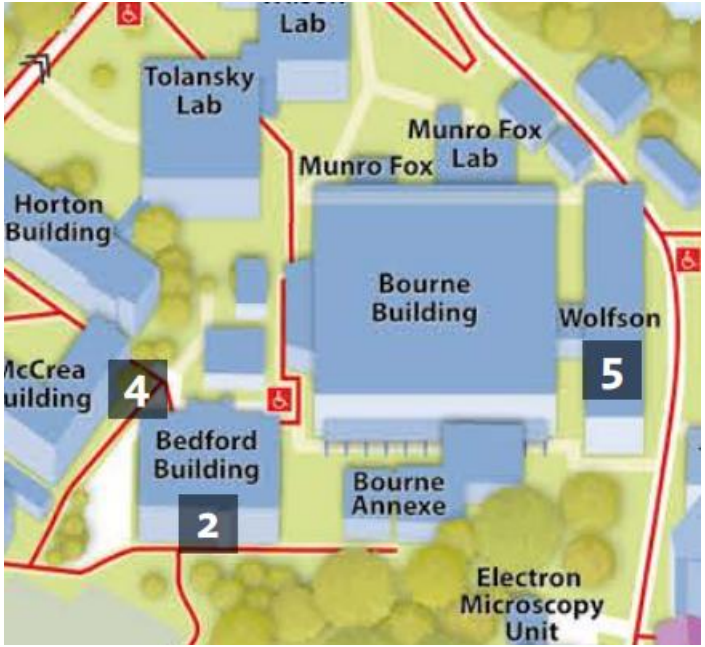
Lecture Schedule

Wednesdays, 10:00 – 13:00

- First lecture: 21st of January
- Last lecture: 1st of April

Room: BOURNE-LT1



Academic Staff / Contact Information

Lecturer: Dr Olga Angelopoulou

- Email: olga.angelopoulou@rhul.ac.uk
- Office: B1-05 (Bedford Building)
- Office hours: Wednesday 1-3pm - By appointment

- use Moodle forums for general organisational or content-related questions
- contact the EPMS School (EPMS-School@rhul.ac.uk) for personal questions regarding module registration, etc.

The Module

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provide the foundations and theoretical underpinnings for an understanding of the way in which data that can subsequently be used as evidence is generated, stored, and transmitted.

methods for the collection and analysis of digital evidence are covered which will not alter the underlying data or potentially trigger destructive mechanisms and which can be reproduced reliably.

the module is to provide an understanding of general and UK legal requirements as well as resulting frameworks for the handling and processing of such evidence.

Learning Outcomes

LO1 - Discuss legal requirements for acquiring, analysing, and presenting digital evidence in a (UK) court of law.

LO2 - Gather host-based evidence from file systems (esp. Microsoft windows-based), email, and audit records.

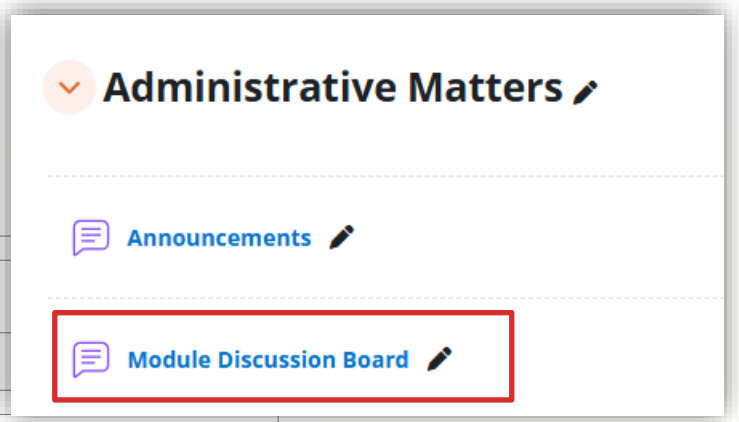
LO3 - Discuss methods and requirements for obtaining network-based evidence.

LO4 - Apply analysis techniques for malware infiltration, exfiltration, and obfuscation mechanisms.

Moodle

Module Summary:
The module is offered as a postgraduate optional module in the MSc in Information Security, and is also offered at the undergraduate level (IY3609 and IY4609) with adaptations in the assessment. The actual content is not differentiated between the module codes.

Module Learning Outcomes:
LO1 - Discuss legal requirements for acquiring, analysing, and presenting digital evidence in a (UK) court of law.
LO2 - Gather host-based evidence from file systems (esp. Microsoft windows-based), email, and audit records.
LO3 - Discuss methods and requirements for obtaining network-based evidence.
LO4 - Apply analysis techniques for malware infiltration, exfiltration, and obfuscation mechanisms.



Indicative Lecture Weekly Plan:

	Theme	Topic
Week 1	Introduction	Introduction to the module Introduction to digital forensics concepts
Week 2		Legislation in DF Collection & acquisition principles and considerations
Week 3	File Systems	Data storage basics and partitioning
Week 4		FIle systems & filesystem metadata
Week 5	Analysis	Host forensics and examination approaches I
Week 6		Host forensics and examination approaches II
Week 7	Specialist themes	Digital forensics in incident response
Week 8		Network forensic principles and practise
Week 9		Malware Analysis for Digital Forensics
Week 10		Mobile forensic principles
Week 11		Revision

Reading List

Disclaimer

This course necessarily is just an introduction to digital forensics and, by design, will touch upon a number areas without being able to cover all or even most of these in depth.

The value derived from the course hence depends very much on individual further reading and research:

- A substantial amount of suggested further reading is provided for each week
- Despite this, it is unlikely that this material can be absorbed in the course of one term
- Focus of the module is necessarily on **foundations** and **selected theoretical aspects**

Assessments

CW1: (Moodle) Quiz – 20% of final mark

- Date: 18th of February
- Duration: 1h

Exam in April/May – 80% of final mark

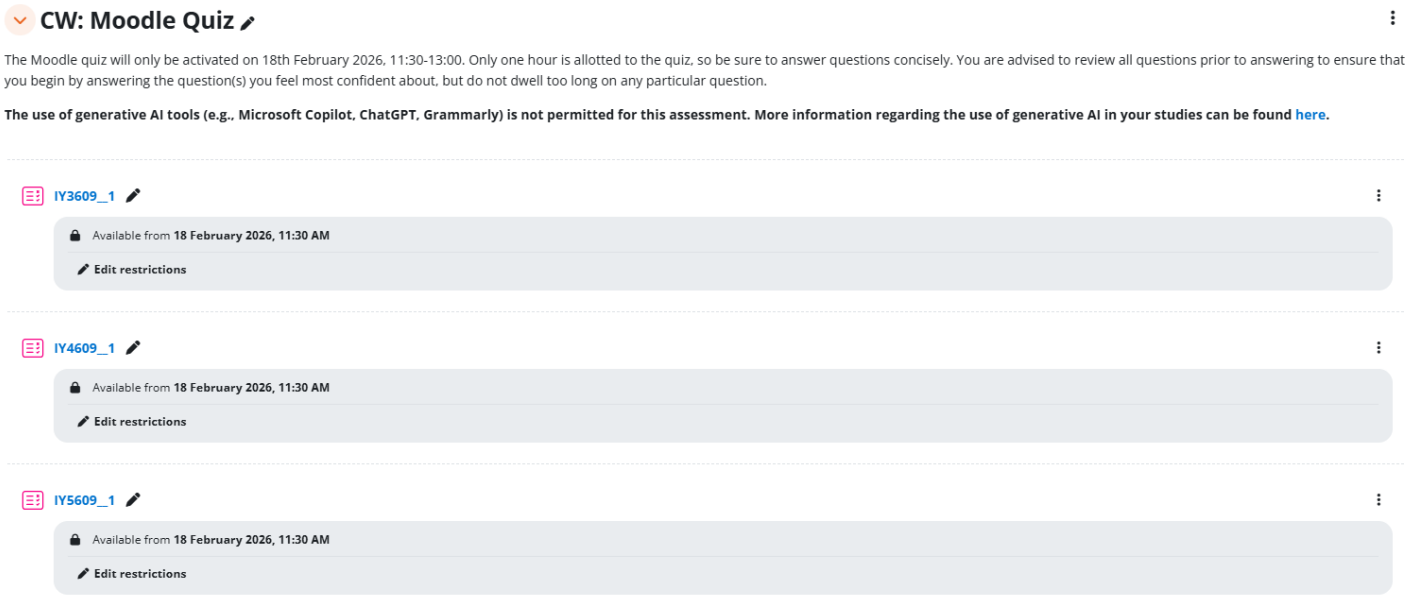
- Format: 2h closed-book exam

Date & Time

- Date: **Wednesday, 18th of February 2025**
- Time:
- 10:00 - 11:00: Regular lecture for one hour
- 11:00 - 11:30: Break
- 11:30 - 13:00: The Moodle quiz will be available. Within this time frame, you have **one hour** to complete the quiz, **D&N adjustments** will be automatically considered.

If you bring a suitable mobile device, you can stay in the lecture hall to work on the quiz. In this case, make sure **beforehand** that your device is **sufficiently charged**, and the **network connection is working**. Alternatively, you can use the break to relocate to one of the PC labs on campus or go home if you live close by.

Moodle Setup



Make sure to select the quiz that corresponds to the **module code for which you are registered** as there will be slight differences in the assessment.

Reading Material

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Basic Textbooks

Digital evidence and computer crime: forensic science, computers and the Internet

Book - by Eoghan Casey - c2011 - 3rd ed - Recommended

File system forensic analysis

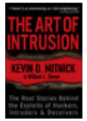
Book - by Carrier, Brian; Safari Tech Books Online. - 2005 - 1st edition - Recommended

Incident response and computer forensics, third edition : Jason T. Luttgens, Matthew Pepe and Kevin Mandia.

Book - by Pepe, Matthew; Kazanciyan, Ryan; Luttgens, Jason T.; Mandia, Kevin - 2014 - Third edition. - Recommended

Implementing digital forensic readiness

Book - by Jason Sachowski - 2019 - Second edition - Further



The art of intrusion : the real stories behind the exploits of hackers, intruders, & deceivers

Book - by Mitnick, Kevin; Simon, William L. - 2005 - 2005; 2005 - Further

Further reading is also provided on Moodle each week!

Questions?

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